

EXETER, United Kingdom

WMO: 038390

Lat: 50.737N

Lon: 3.404W

Elev: 102

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	27.3	21.9	16.5	27.0	24.5	18.7	30.4	29.8	51.2	27.7	50.7	2.9	30	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	77.1	64.8	73.8	63.2	71.6	62.4	66.6	74.3	65.0	71.6	63.6	69.3	9.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	89.8	69.3	62.6	85.2	67.4	61.1	80.7	65.9	31.1	74.1	30.0	71.6	29.0	69.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	21.5	19.0		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	42.1	42.9	44.4	47.8	53.3	58.8	62.0	61.3	57.7	53.0	46.8	43.0
	DBStd	8.93	6.43	5.55	5.12	4.77	4.61	4.14	4.12	4.03	4.23	5.20	6.30	7.63
	HDD50	1149	252	203	182	96	18	1	0	0	2	30	132	233
	HDD65	5098	710	618	639	517	364	191	111	128	221	371	546	682
	CDD50	1563	7	5	8	29	120	264	371	349	233	124	36	17
	CDD65	38	0	0	0	0	1	5	18	12	2	0	0	0
	CDH74	257	0	0	0	0	6	43	144	60	3	1	0	0
	CDH80	24	0	0	0	0	0	2	15	7	0	0	0	0
	WSAvg	8.4	9.7	9.5	9.3	9.2	8.0	7.8	7.8	7.5	7.6	8.2	8.2	7.8
	PrecAvg	31.9	3.9	2.9	2.5	2.1	2.2	1.9	1.8	2.2	2.5	3.3	3.0	3.6
(15) Precipitation	PrecMax	50.6	8.8	6.4	5.2	5.6	5.8	4.7	4.4	5.1	7.3	12.2	8.3	7.7
	PrecMin	20.6	0.4	0.1	0.0	0.1	0.1	0.0	0.6	0.4	0.5	0.3	0.7	0.6
	PrecStd	5.7	2.0	1.8	1.1	1.5	1.4	1.2	1.0	1.1	1.8	2.6	1.9	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	55.8	61.7	69.8	74.5	78.6	82.3	81.0	74.8	66.8	61.4	58.0
		MCWB	54.4	52.0	53.8	58.5	62.4	65.6	66.6	67.8	65.1	60.9	58.9	55.0
	2%	DB	55.6	54.3	57.4	62.4	69.4	74.8	78.2	75.5	70.2	64.3	59.2	56.7
		MCWB	53.1	51.0	51.7	54.9	60.0	63.4	65.4	64.2	62.6	59.9	56.5	54.5
	5%	DB	53.8	53.2	55.3	59.0	65.9	71.5	74.9	72.0	68.1	62.6	57.6	55.3
		MCWB	51.7	50.1	50.9	52.2	58.5	62.0	63.8	62.8	61.3	59.0	55.0	53.1
	10%	DB	52.0	51.6	53.5	56.3	62.6	68.2	71.5	69.8	66.1	61.0	56.3	53.6
		MCWB	49.8	48.7	49.7	50.9	56.4	60.1	62.4	61.8	60.1	57.7	53.8	51.3
	0.4%	WB	55.3	53.6	54.7	58.9	64.1	67.3	68.2	69.1	66.7	62.8	59.6	55.9
		MCDB	56.9	55.5	59.5	66.9	71.4	75.0	78.3	77.8	72.0	65.3	61.0	57.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.7	52.0	53.0	55.4	61.0	64.6	66.4	66.2	64.4	61.2	57.2	54.8
		MCDB	55.4	54.2	56.5	61.2	67.3	71.6	75.7	72.4	68.9	63.3	58.7	56.3
	5%	WB	52.1	50.6	51.8	53.3	59.1	62.8	64.9	64.4	62.9	59.8	55.7	53.4
		MCDB	53.8	52.4	54.8	57.4	64.8	69.5	72.5	69.7	66.5	61.9	57.3	55.0
	10%	WB	50.3	49.3	50.4	51.9	57.0	61.0	63.4	63.1	61.3	58.5	54.2	51.7
		MCDB	51.8	51.0	52.8	55.5	61.7	67.2	69.7	68.0	64.7	60.7	56.2	53.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.6	14.5	15.8	16.4	16.4	16.8	16.2	15.9	13.2	12.1	12.0
		MCDBR	11.4	11.5	17.3	20.3	21.0	22.3	22.4	19.4	18.0	13.3	10.1	9.0
		MCWBR	11.1	9.3	12.5	13.6	13.2	12.8	11.5	10.6	11.6	9.8	9.1	8.5
	5% WB	MCDBR	11.5	10.9	15.9	17.5	18.8	18.7	19.0	16.3	15.4	11.8	10.4	9.4
		MCWBR	11.5	9.6	12.5	12.6	12.5	11.7	10.9	10.1	11.0	9.6	9.8	9.3
(40) Clear-Sky Solar Irradiance	taub		0.335	0.349	0.383	0.399	0.402	0.393	0.389	0.379	0.371	0.360	0.333	0.325
	taud		2.339	2.357	2.266	2.250	2.279	2.341	2.357	2.405	2.415	2.411	2.393	2.314
	Ebn at Noon		215	246	257	267	271	274	273	271	259	239	215	197
	Edh at Noon		21	27	35	39	40	38	37	33	30	25	20	19
(44) All-Sky Solar Radiation	RadAvg		257	490	874	1370	1660	1800	1705	1424	1076	602	322	204
	RadStd		23	55	111	128	87	180	130	114	70	46	31	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page